DVT — Deploy Via Torrent: Manual

English | 한국어

DVT restores a Windows image to many PCs at once over the local network. Each PC boots a small WinPE image, downloads the backup image with BitTorrent (aria2), and applies it with wimlib — so every PC that has finished downloading also uploads to the others, and adding more PCs does not slow the deployment down the way a single file server would.

A client PC can start DVT in two ways:

- **Client local boot** — `C:\temp\boot\make_boot.bat` runs in the client's Windows, prepares the WinPE image for that PC and restarts into it. There is no need to go to the client PC to enter anything.
- **USB / CD media** — `02_make_media.bat` makes a boot USB drive or ISO. Use it for PCs that cannot start Windows or have no Windows installed. The operator enters the IP address and the disk at each PC.

Everything is plain Windows batch. The scripts that run inside WinPE use only tools that WinPE ships with (no WMIC, WMI, PowerShell or findstr), plus aria2c, wimlib-imagex and NirCmd.

> **Warning**
>
> DVT **erases a whole disk** on every client PC it runs on. Read Choosing the disk to erase and test on a spare PC before using it in a lab or office.

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How it works

	Client local boot	USB / CD media
Build PC	`01_make_pe.bat` → `_output\boot\` folder	`02_make_media.bat` → ISO or USB drive
Client PC, before WinPE	`C:\temp\boot\make_boot.bat` in Windows (once per PC)	nothing — boot from the media
WinPE main script	`bats\pe\start_dvt.bat`	`bats\Media\pe\start_media.bat`
IP address	detected in Windows (static IP → `ip.txt`)	static IP from `ip.txt` (operator types the last octet) or DHCP, order by `DHCP_ENABLE`
Disk to erase	detected (disk ID, then the RAW-volume fallback)	typed by the operator
Failure before erasing	reboots into the old Windows	waits for a key, then reboots
WinPE settings	`bats\pe\config.cmd` (inside `boot.wim`)	`bats\Media\pe\config.cmd` (also on the media, editable on a USB drive)

Both ways check the gateway, the seeder and the disk space **before** anything is erased, and both use the same torrent, aria2 profiles, download and restore (`restore.bat`, `setpc.bat`). The WinPE scripts that both use are kept once, in `bats\common`.

Client local boot

```
[1] Build PC ── 01_make_pe.bat
      ADK winpe.wim + DVT scripts + aria2c, wimlib
      → _output\boot\ folder (boot.wim, boot.sdi, host scripts)
         │
         │  copy _output\boot\ to C:\temp\boot on the client
         ▼
[2] Client PC (Windows) ── C:\temp\boot\make_boot.bat
      • detect the IP address (and static DNS) and link speed
      • read the disk ID of the Windows disk
      • export the network drivers
      • add drivers, ip.txt, torrent, aria2.conf, target_disk.txt
        to boot.wim
      • suspend BitLocker for one restart (if it is on)
      • add a one-time boot entry
         │
         │  reboot
         ▼
[3] Client PC (WinPE) ── start_dvt.bat
      • find the target disk (disk ID, then fallback)
      • configure the network
      • check the gateway and seeder
      • check the disk space           ← a failure up to here reboots into
                                            the old Windows, disk untouched
      • erase the disk, create partitions
      • download by torrent (aria2c)
         │
         │  download complete (aria2c keeps seeding)
         ▼
    restore.bat
      • apply the WIM, drivers, bcdboot, setpc.bat
      • record version and image in HKLM\SOFTWARE\DVT
         │
         ▼
    reboot into the restored Windows
```

USB / CD media

```
[1] Build PC ── 02_make_media.bat
      source\winpe.wim (clean ADK copy)
      + pe_tools, drivers, nircmd
      + bats\Media\pe and bats\common  (only these scripts)
      → _output\media\boot.wim
      + source\adk_media, bats\Media\config, bats\Media\pe\config.cmd
      → ISO (oscdimg) or USB drive
         │
         │  boot from the USB drive / CD
         ▼
[2] Client PC (WinPE) ── startnet.cmd
      • copy <media>\config\* → X:\restore
         ▼
    start_media.bat
      • network order (DHCP_ENABLE: static first / DHCP first)
      • show the disks, enter "150.0" (last IP octet.disk number)
      • check the gateway
      • link.exe → aria2.conf for the link speed
      • check the seeder
      • check the disk space          ← nothing is erased if a step up to
                                           here fails
      • erase the disk, create partitions
      • choose the download drive, download by torrent (aria2c)
         │
         │  download complete (aria2c keeps seeding)
         ▼
    restore.bat
      • apply the WIM, drivers, bcdboot
      • move files, setpc.bat, record in HKLM\SOFTWARE\DVT
         │
         ▼
    reboot (remove the USB drive / CD)
```

The media image does **not** use `_output\boot\boot.wim` or the `bats\pe` folder: `X:\restore` contains only the tools of `pe_tools` and the files of `bats\Media\pe` and `bats\common`.

Features

- **BitTorrent distribution** with aria2c; the restore starts as soon as the download is complete, and the PC keeps seeding while the image is applied.
- **Safety checks before erasing**: target disk, network, gateway, a real seeder download and the disk space are verified first. Nothing is erased when one of them fails.
- **Target disk detection** by the disk ID (GPT disk GUID / MBR signature) read in the running Windows, with an explicit override and a careful fallback for instant recovery programs. Anything ambiguous is an error. On the USB / CD media the operator chooses the disk; the boot media itself cannot be chosen.

- **BitLocker**: the disk ID works on an encrypted Windows drive, and the protection is suspended for one restart, so a PC that returns to Windows after a failed check does not ask for the recovery key.
- **UEFI and legacy BIOS**: GPT or MBR layout and the `bcdboot` firmware type are chosen automatically. The ISO and the USB drive boot on both.
- **Static IP or DHCP**: a static address (with its static DNS servers) used by Windows is carried into WinPE and applied to whichever adapter reaches the gateway. The media tries a static IP or DHCP first (`DHCP_ENABLE`) and falls back to the other.
- **Network drivers from the running PC** are exported and injected into WinPE, so NICs that WinPE does not support out of the box still work.
- **Link-speed profiles**: separate aria2 settings for 100 Mbps and 1 Gbps links.
- **Post-restore tasks**: extra files are copied to `C:\temp` and an optional `setpc.bat` runs once at the first Windows start.
- **Restore record**: the DVT version, the image, the date and more are written to `HKLM\SOFTWARE\DVT` of the restored Windows (see Version).
- **One-time boot entry** that is removed automatically.
- **USB / CD media** for PCs without a working Windows; its settings can be changed on the USB drive without rebuilding.
- **Large IP display** on the client so you can identify a PC from across the room.

Requirements

Build PC

- Windows 10/11 x64, administrator rights
- Windows ADK and the **Windows PE add-on** (amd64). The USB / CD media also needs the **Deployment Tools** of the ADK, on the first run only (see below).
- `aria2c.exe` and `wimlib-imagex.exe` (with its DLL), placed in `pe_tools\` or `system32\`
- `nircmd.exe` (x64), placed in `nircmd-x64\`
- USB / CD media only: a 64-bit `link.exe` in `bats\Media\pe`. It is used as it is (nothing is compiled); `02_make_media.bat` stops when it is missing.

`02_make_media.bat` stores the ADK `Media` folder in `source\adk_media`, and oscdimg, the boot sector files and bootsect in `source\adk_tools`. Later runs use only the stored copies (and `source\winpe.wim`, shared with `01_make_pe.bat`). `01_make_pe.bat` does not have to run first.

Client PCs

- Client local boot: Windows 10/11 x64 with administrator rights (to run `make_boot.bat`)
- USB / CD media: a PC that can boot from USB or CD (UEFI or BIOS)
- Wired network connection to the deployment network

Deployment network

- A BitTorrent **tracker** reachable from the clients. DHT, peer exchange and local peer discovery are disabled in the aria2 profiles, so peers are found through the tracker only.
- At least one **seeder** that has the complete `DVT_backup` folder.
- qBittorrent can be both the seeder and the tracker. See qBittorrent setup.

Repository layout

```
dvt/
├─ 01_make_pe.bat           build the WinPE image (build PC)
├─ 03_make_test_pe.bat      build the test WinPE image (build PC)
├─ 02_make_media.bat        build the USB / CD media: ISO, new USB drive,
│                           or update a USB drive (build PC)
├─ qbittorrent-setup.en.md  qBittorrent seeder and tracker setup manual
├─ qbittorrent-setup.ko.md  the same manual in Korean
├─ doc/                     manual.md / manual.ko.md (README + qBittorrent
│  │                        manual in one file) and their PDFs
│  └─ qbittorrent.ini/      qBittorrent settings files: qbittorrent.ini.1G,
│                           qbittorrent.ini.100
├─ bats/                    all scripts
│  ├─ common/                  WinPE scripts shared by the client local boot,
│  │  │                        the test image and the USB / CD media
│  │  │                        (copied to X:\restore after pe/ or Media/pe/)
│  │  ├─ version.cmd           the DVT version (the only place it is set)
│  │  ├─ restore.bat           apply the image, boot files, record, clean-up
│  │  ├─ start_restore.bat     called by aria2c when the download is complete
│  │  ├─ check_space.bat       disk space check before erasing
│  │  └─ ...                   check_gateway, check_seeder, get_firmware,
│  │                           get_ip, get_target, make_disk, ping_host,
│  │                           set_ip, wait (see Script reference)
│  ├─ host/                    Windows-side scripts (client PC)
│  │  ├─ config.cmd            settings for the host scripts and 01_make_pe.bat
│  │  ├─ make_boot.bat         prepare a client PC and restart into WinPE
│  │  ├─ get_host_ip.bat       IP address, static DNS and static/DHCP detection
│  │  ├─ get_link_speed.bat    link speed -> aria2 profile
│  │  ├─ get_disk_id.bat       disk ID of the Windows disk
│  │  ├─ suspend_bitlocker.bat BitLocker suspended for one restart
│  │  ├─ export_net_drivers.bat
│  │  ├─ prepare_mount_dir.bat
│  │  ├─ set_bcd.bat           one-time WinPE boot entry
│  │  └─ show_ip.bat           IP address in large digits
│  ├─ pe/                      WinPE scripts of the client local boot
│  │  ├─ config.cmd            settings for the WinPE scripts
│  │  ├─ startnet.cmd          WinPE entry point
│  │  ├─ start_dvt.bat         main flow
│  │  ├─ get_disk.bat          disk to erase (disk ID, fallback)
│  │  ├─ aria2.conf.100        aria2 profile for 100 Mbps links
│  │  └─ aria2.conf.1G         aria2 profile for 1 Gbps links
│  ├─ Media/                   USB / CD media
│  │  ├─ config.cmd            settings for 02_make_media.bat (loads host\config.cmd)
│  │  ├─ link.au3              source of link.exe (reference only)
│  │  ├─ config/               copied to <media>\config
│  │  │  ├─ DVT_backup.torrent
│  │  │  ├─ aria2.conf.100
```

```
|  |  |  ├─ aria2.conf.1G
|  |  |  ├─ ip.txt             static IP (see ip.txt.example)
|  |  |  └─ ip.txt.example
|  |  ├─ pe/                   WinPE scripts of the media only -> X:\restore
|  |  |  ├─ config.cmd         settings for the media WinPE scripts
|  |  |  |                     (also copied to <media>\config)
|  |  |  ├─ startnet.cmd       find the media, copy config\, start start_media.bat
|  |  |  ├─ start_media.bat    main flow
|  |  |  ├─ get_input.bat      disk list, IP last octet and disk number
|  |  |  ├─ getdiskinfo.bat    disk list
|  |  |  └─ link.exe           aria2.conf by link speed (64-bit, required)
|  |  └─ test/                 test image scripts (see Test image)
|  |     ├─ test_config.cmd    FTP server and test settings
|  |     ├─ startnet.cmd       test WinPE entry point
|  |     ├─ start_test.bat     test flow: network, gateway, seeder, upload
|  |     └─ upload_ftp.bat     FTP upload (curl.exe or ftp.exe)
|  |
|  ── not in the repository, provided by you ──
├─ pe_tools/               aria2c.exe, wimlib-imagex.exe, DLLs -> X:\restore
├─ system32/               optional files -> X:\Windows\System32
├─ drivers/                optional drivers added to every WinPE image
├─ nircmd-x64/nircmd.exe
|
|  ── created by the build scripts ──
├─ source/                 stored copies of ADK files (first run only)
|  ├─ winpe.wim, boot.sdi  clean copies (01_make_pe.bat, 02_make_media.bat)
|  ├─ adk_media/           the ADK WinPE "Media" folder (02_make_media.bat)
|  └─ adk_tools/           oscdimg.exe, etfsboot.com, bootsect.exe,
|                          efisys.bin (= ADK efisys_noprompt.bin, renamed)
└─ _output/                build results
   ├─ boot/                01_make_pe.bat: boot.wim + boot.sdi + host scripts
   |                       + version.cmd
   |                       (copy this folder to the client as C:\temp\boot)
   ├─ boot_test/           03_make_test_pe.bat (copy to the client as C:\temp\boot_test)
   └─ media/               02_make_media.bat: boot.wim of the media, DVT_media.iso
```

Quick start — client local boot

1. Build the WinPE image (build PC)

1. Install the Windows ADK and the WinPE add-on.
2. Put `aria2c.exe`, `wimlib-imagex.exe` and its DLL in `pe_tools\`, and `nircmd.exe` in `nircmd-x64\`.
3. Optionally add drivers to `drivers\` (e.g. storage drivers for new hardware, or the WinPE driver for Realtek NICs).
4. Review `bats\pe\config.cmd` (see Configuration).
5. Run `01_make_pe.bat`. It asks for elevation if needed.

The result is the `_output\boot\` folder: `boot.wim`, `boot.sdi`, a copy of every file in `bats\host\` and `version.cmd`. This folder is everything a client PC needs besides the torrent.

> The mount and work folders are on the local system drive (`C:\WinPE_LocalBuild`). Do not move them into a folder synced by OneDrive, Dropbox or similar: sync clients can bring back deleted files after the unmount.

2. Prepare the torrent and the seeder

See Preparing the torrent. Start the tracker and the seeder (with qBittorrent, see qBittorrent setup).

3. Prepare each client PC

Copy the `_output\boot\` folder from the build PC to `C:\temp\boot`, and the torrent to `C:\temp`:

```
C:\temp\
├─ boot\
│  ├─ boot.wim
│  ├─ boot.sdi
│  ├─ make_boot.bat
│  ├─ config.cmd
│  └─ ...              (the other host scripts)
└─ DVT_backup.torrent
```

Run `C:\temp\boot\make_boot.bat`. It:

1. detects the IP address (a static address is saved to `ip.txt`: ip, subnet, gateway and the static DNS servers, one per line),
2. reads the link speed and picks `aria2.conf.100` or `aria2.conf.1G`,
3. reads the disk ID of the Windows disk,
4. exports the drivers of the connected network adapters,
5. mounts `boot.wim` and adds the drivers, `ip.txt`, the torrent, `aria2.conf` and `target_disk.txt` (the disk ID),
6. suspends BitLocker on the Windows drive for one restart, if it is on,
7. adds a one-time WinPE boot entry,
8. shows the IP address in large digits and restarts after a key press.

4. Let it run

The PC boots into WinPE and restores itself. The console color tells you the state:

Console	Meaning
Light gray background	Checking, partitioning, downloading
Green background (new "DVT Restore" window)	Restore running, then completed and rebooting
Red background	Failed — read the message on screen

If a check fails **before** the disk is erased, the PC reboots back into its old Windows after a few seconds. If something fails **after** the disk was erased, DVT waits for a key so you can read the error.

Quick start — USB / CD media

1. Prepare the torrent and the seeder (see Preparing the torrent and qBittorrent setup).
2. Put the files in `bats\Media\config`: `DVT_backup.torrent`, `aria2.conf.100`, `aria2.conf.1G`, and `ip.txt` for a static IP (see Network).
3. Review `bats\Media\pe\config.cmd` (see Configuration).
4. Run `02_make_media.bat` in the build folder as administrator and choose:
   - `1` ISO → `_output\media\DVT_media.iso`, burn it to a CD / DVD
   - `2` USB → the selected **USB** disk is erased and made bootable (FAT32 on the whole disk; `USB_FS` and `USB_PART_MB` in `bats\Media\config.cmd`)
   - `3` update → copies new files to an existing DVT USB drive

   `02_make_media.bat iso`, `02_make_media.bat usb` and `02_make_media.bat update` skip the menu.
5. Boot the client PC from the media, read the disk list, and type `<last octet>.<disk>` (e.g. `150.0`) for a static IP, or only the disk number for DHCP, then `Y`.
6. When the restore is done the PC reboots in 15 seconds — remove the media.

> **Warning**
>
> The disk number you type is **erased**. Check the disk list carefully. The disk of the boot USB drive cannot be selected.

Failures on the media always wait for a key, because an operator is present.

Test image

`03_make_test_pe.bat` builds a **test** WinPE image that checks a client PC end to end without touching any disk. Use it to find network, driver and seeder problems before the real restore. It is started the same way as the client local boot (`make_boot.bat`).

In WinPE the test image:

1. sets up the network exactly like the restore image (static IP from `ip.txt`, otherwise DHCP) and checks the gateway,
2. runs the seeder check (downloads `dvt_check.txt` only, into memory) and the disk space check (report only, it never stops the test),
3. writes a report `<ip>.txt` (DVT version, IP, gateway, adapter, link profile, seeder, firmware, detected target disk and how it was found, disk space, model, `diskpart list disk`, `detail disk` of the target disk, `ipconfig /all`),
4. uploads it to the FTP server and reboots back into Windows.

The erase and restore scripts (`make_disk.bat`, `restore.bat`, `start_restore.bat`, `start_dvt.bat`) are **not** copied into the test image.

Steps:

1. Set the FTP server in `bats\test\test_config.cmd`: `FTP_HOST`, `FTP_PORT`, `FTP_USER`, `FTP_PASS` and, optionally, `FTP_DIR` (relative to the login folder). Do not use `!` or `"` in these values.
2. Run `03_make_test_pe.bat` on the build PC. The result is the `_output\boot_test\` folder.
3. On the client, copy `_output\boot_test\` to `C:\temp\boot_test`, put the torrent in `C:\temp` and run `C:\temp\boot_test\make_boot.bat`. `C:\temp\boot` and `C:\temp\boot_test` can be on the client at the same time: `make_boot.bat` always uses the `boot.wim` in its own folder.

Screen	Meaning
Green "PASS"	Gateway and seeder OK, `<ip>.txt` uploaded, rebooting
Red "TEST FAILED"	A check or the upload failed. Reboots after `FAIL_REBOOT_SEC` (`TEST_FAIL_REBOOT=1`, default); `TEST_FAIL_REBOOT=0` waits for a key

With `TEST_UPLOAD_FAIL=1` a failed test also uploads `<ip>_FAIL.txt` when an IP address was set. The FTP client is `curl.exe` (passive mode, creates `FTP_DIR`) or, as a fallback, `ftp.exe` (active mode only, `FTP_DIR` must exist). WinPE contains neither: `03_make_test_pe.bat` copies them from the build PC unless you put them in `pe_tools\` or `system32\`. The FTP settings are built into the image, so rebuild it after changing them.

Use it with a PC management tool such as Veyon

DVT needs two things on every client before WinPE takes over: the `C:\temp\boot` folder with the torrent, and one run of `make_boot.bat`. Doing that PC by PC is the slowest part of a deployment. A classroom / PC management tool like Veyon (free, open source) lets you do it for the whole room from one screen:

DVT step	Without a management tool	With Veyon
Turn the PCs on	Walk to every PC	**Power on** (Wake-on-LAN) for all PCs
Copy `C:\temp\boot` and the torrent	USB stick at every PC	**File transfer** to all PCs, then **Run program** to put the files in `C:\temp`
Run `make_boot.bat`	Log in and start it on every PC	**Run program** on all PCs at once
Check IP and link speed	Read every monitor	`make_boot.bat` shows the IP in large digits and colors the console by link speed — visible at a glance in Veyon's **monitoring** view
Restart into WinPE	Press a key on every PC	**Remote control**, or **Restart** all PCs
Confirm the restore	Walk around again	Restored PCs show up in Veyon again (if the Veyon service is in the image)

The same works for the **test image**: run it on the whole room first, collect the `<ip>.txt` reports from FTP, fix the failing PCs, then deploy for real.

Things to keep in mind:

- `make_boot.bat` needs **administrator rights**. Approve the elevation prompt through Veyon's remote control, or run it with an account that is an administrator.
- Veyon puts transferred files in the user's profile folder by default; move them to `C:\temp` (e.g. with a small script run through **Run program**).
- While a PC is in **WinPE** the Veyon service is not running, so Veyon cannot show it. Read the PC's own screen, or use the test image's FTP reports.

- Put the Veyon service into the **backup image** (and, if needed, set it up in `setpc.bat`), so every restored PC can be managed right away.

Preparing the torrent

The torrent must contain a single top folder named `DVT_backup` (`DL_NAME`):

```
DVT_backup\
├─ backup.wim        the Windows image (index WIM_INDEX, default 1) required
├─ dvt_check.txt     small file (a few KB) for the seeder check     required
├─ drivers\          drivers added to the restored Windows          optional
└─ files\            copied to C:\temp of the restored Windows      optional
   └─ setpc.bat      runs once at the first Windows start           optional
```

- Create `dvt_check.txt` **before** creating the torrent. WinPE downloads only this file to prove that a seeder is reachable before it erases anything.
- Put the tracker URL in the torrent. Alternatively set `SEEDER_EXTRA_OPTS=--bt-tracker=http://...` in `bats\pe\config.cmd` or `bats\Media\pe\config.cmd` (this is used for the seeder check only; the main download reads the torrent and `aria2.conf`).
- Save the torrent as `DVT_backup.torrent`: in `C:\temp` of each client for the client local boot, in `bats\Media\config` for the media.
- `backup.wim` may hold several images. `WIM_INDEX` in `bats\pe\config.cmd` and `bats\Media\pe\config.cmd` selects the one to apply (default `1`). List them with `wimlib-imagex info backup.wim`. A wrong index stops the restore with a message before the image is applied.
- When the client uses a static IP, `ip.txt` (ip, subnet, gateway, and the DNS servers when known) is also copied to `C:\temp` of the restored Windows, so `setpc.bat` can set the address again.

> **Warning**
>
> `dvt_check.txt` must **not be an empty 0-byte file.** There is no data to download, so aria2c "completes" it without reaching any seeder and the check would prove nothing. `check_seeder.bat` therefore fails the check, before the disk is erased, when `dvt_check.txt` in the torrent is 0 bytes ("The seeder check file in the torrent is empty"). Always put a file with content (a few KB) in the folder before you create the torrent. If you change the file, create the torrent again.

qBittorrent setup (seeder and tracker)

One qBittorrent PC can serve the image (seeder) and broker the peer list (embedded tracker) at the same time. The settings differ between a 100 Mbps or slower network and a 1 Gbps or faster network, and each is split into required and recommended items.

> **Important**
>
> This manual is written for the **standard installer of qBittorrent 5.3.0** (`qbittorrent_5.3.0_x64_setup.exe`, libtorrent 2.1). From 5.3.0 the standard installer is based on libtorrent 2.1.x. The lt12 installer (`qbittorrent_5.3.0_lt12_x64_setup.exe`) uses libtorrent 1.2 and shows a few different items; see lt12 build (libtorrent 1.2).

> **Note**
>
> The qBittorrent 5.3.0 release notes say that the libtorrent 2.1 backend has worse read performance than 1.2, which can lower upload speed on a heavy seeder. If the seeder cannot fill a 1 Gbps or faster link after the settings in **1 Gbps or faster** are applied, try the lt12 build.

Overview

In DVT, one qBittorrent PC acts as both the **seeder** (serves the source image) and the **embedded tracker** (brokers the peer list). Clients booted into WinPE download the image with aria2 and exchange pieces with each other.

- At 100 Mbps or slower, the **network bandwidth** is the bottleneck. The goal is to save seeder upload and let clients share pieces among themselves.
- At 1 Gbps or faster, the seeder's **disk, CPU and socket buffers** are the bottleneck. The goal is to raise buffers and I/O threads so the link stays full.

How to use: apply the common settings in **Common settings** first, pick your environment with **Choosing the environment**, then apply **100 Mbps or slower** or **1 Gbps or faster** on top. Required items are those whose absence makes deployment fail or stall; recommended items improve stability and efficiency. Names follow the qBittorrent 5.3.0 **English UI**: settings are under **Tools → Options**, torrent creation under **Tools → Torrent Creator**. Wording may differ slightly between versions. For the Korean UI, see the Korean edition of this manual (manual.ko.md). Instead of changing each option, you can also replace the settings file with a ready-made one.

Common settings (both environments)

Turn on the embedded tracker, lock the protocol to TCP, and create torrents in **V1 format**. aria2 supports neither µTP nor BitTorrent v2.

Required

Page	Setting	Value	Why
Advanced	Enable embedded tracker	Checked	qBittorrent acts as the tracker
Advanced	Embedded tracker port	9000 (default)	Port used in the torrent's tracker URL
Connection	Peer connection protocol	TCP	aria2 does not support µTP
Connection	Port used for incoming connections	Fixed port (e.g. 6881)	Keeps firewall rules stable
BitTorrent	Torrent Queueing	Unchecked	Keeps the deployment torrent out of the queue
BitTorrent	Seeding Limits (ratio / seeding time)	All unchecked	Seeding must not stop mid-deployment
Torrent Creator	Torrent format	**V1**	aria2 cannot download V2 or Hybrid torrents
Torrent Creator	Tracker URLs	`http://<SEEDER_IP>:9000/announce`	Embedded tracker address; use the seeder's exact LAN IP

Page	Setting	Value	Why
Torrent Creator	Start seeding immediately	Checked	Seeding starts as soon as the torrent is created
Torrent Creator	Piece size	See **100 Mbps or slower** or **1 Gbps or faster**	Balances piece count and hashing load

Firewall (seeder PC, elevated Command Prompt) — required

```
netsh advfirewall firewall add rule name="DVT qBittorrent tracker" dir=in action=allow protocol=TCP localport=9000
netsh advfirewall firewall add rule name="DVT qBittorrent peer" dir=in action=allow protocol=TCP localport=6881
```

If you change either port, change `localport` to match. Add these port rules even if a program-level rule exists, because a network on the Public profile can still be blocked.

Recommended

Page	Setting	Value	Why
Advanced	Network interface / Optional IP address to bind to	Deployment LAN adapter / seeder IP	Effectively required on multi-NIC seeders; prevents traffic leaving on the wrong network
Advanced	Enable port forwarding for embedded tracker	Unchecked	LAN only
Connection	Use UPnP / NAT-PMP port forwarding from my router	Unchecked	LAN only
BitTorrent	Enable DHT / Enable Peer Exchange (PeX)	Unchecked	Peers are found through the tracker; avoids outside exposure
BitTorrent	Enable Local Peer Discovery	Unchecked (if the tracker works)	Redundant with the tracker; a fallback when troubleshooting
BitTorrent	Encryption mode	Allow encryption	Compatible with aria2; forcing it costs CPU
BitTorrent	Enable anonymous mode	Unchecked	Peer identification is needed
Speed	Schedule the use of alternative rate limits	Unchecked	Prevents sudden slowdowns during deployment
Behavior	Power Management: Inhibit system sleep when torrents are seeding	Checked	Keeps the seeder awake during deployment
Torrent Creator	Private torrent	Checked	DHT and PeX stay off for this torrent regardless of global settings

100 Mbps or slower

Keep seeder upload slots low so each peer completes pieces quickly and passes them on to other clients. A 100 Mbps link tops out around 12 MB/s, so with more than 10 clients the seeder alone drops to roughly 1 MB/s per client.

Required

Page	Setting	Value	Why
Connection	Maximum number of upload slots per torrent	8	Fewer slots mean faster per-peer speed and quicker piece completion for redistribution
Connection	Global maximum number of upload slots	20	Keeps the per-torrent slots from being capped globally
Torrent Creator	Piece size	4 MiB	Smaller pieces let clients start exchanging sooner

Recommended

Page	Setting	Value	Why
Speed	Global Rate Limits → Upload	11000 KiB/s (about 90 Mbps)	Leaves headroom for remote management and tracker replies
Speed	Apply rate limit to peers on LAN	Checked when using the limit above	Otherwise LAN peers are not limited
Speed	Apply rate limit to transport overhead	Checked when using the limit above	Limits by actual link usage
Connection	Global maximum number of connections	200	Comfortably above the peer count
Connection	Maximum number of connections per torrent	100	
Advanced	Upload slots behavior	Fixed slots	Locks the slot count to the values above
Advanced	Upload choking algorithm	Round-robin	Spreads pieces evenly across clients
Advanced	Disk IO read mode	Enable OS cache	Fewer disk re-reads when the same piece goes to many peers

Leave all other Advanced settings at their defaults. Disk and CPU are not the bottleneck here.

1 Gbps or faster

Remove all rate limits and raise upload slots, send buffers and disk I/O so the seeder fills the link. 1 Gbps is about 119 MB/s, which default send buffers and an HDD often cannot sustain.

Required

Area	Setting	Value	Why
Hardware	Image storage	SSD or faster (NVMe for 10 Gbps)	An HDD cannot keep up with random reads from many peers
Hardware	Seeder NIC and switch uplink	1 Gbps or faster	A 100 Mbps uplink means a mixed network (**Choosing the environment**)
Speed	Global Rate Limits → Upload / Download	0 (unlimited)	
Connection	Global maximum number of upload slots	Unchecked (unlimited)	Bandwidth allows sending to every peer at once
Connection	Maximum number of upload slots per torrent	50 or unchecked	

Area	Setting	Value	Why
Advanced	Send buffer watermark	5120 KiB (default 500)	Prevents send buffer starvation on fast links
Advanced	Send buffer low watermark	1024 KiB (default 10)	
Advanced	Send buffer watermark factor	150 % (default 50)	
Torrent Creator	Piece size	8 MiB (16 MiB for images over 50 GB)	Fewer pieces and less hashing load

Recommended

Area	Setting	Value	Why
Hardware	Seeder RAM	16 GB or more	Pieces stay in the OS cache, reducing disk reads
NIC properties	Energy Efficient Ethernet / Green Ethernet	Disabled	Reduces speed fluctuation during high-speed transfers
NIC properties	Jumbo Packet	Disabled (keep default)	Every WinPE driver and switch must support it, otherwise connections break
Connection	Global maximum number of connections	500	
Connection	Maximum number of connections per torrent	300	
Advanced	Upload choking algorithm	Round-robin	
Advanced	Asynchronous I/O threads	16 (default 10)	More concurrent read requests
Advanced	Hashing threads	Half the CPU core count	Faster torrent creation and rechecks
Advanced	Physical memory (RAM) usage limit	4096 MiB or more (to fit RAM)	Room for caching
Advanced	Disk queue size	Default (102400 KiB)	5.3.0 raised the default to 100 MiB for libtorrent 2.1; do not lower it (lt12 build: see below)
Advanced	Disk IO type	Default	Other choices: Memory mapped files, POSIX-compliant, Simple pread/pwrite, Pread/pwrite. If upload falls short or memory use is excessive, compare with Pread/pwrite (restart required)
Advanced	Disk IO read mode	Enable OS cache	Repeatedly requested pieces are served from RAM
Advanced	Socket backlog size	100 (default 30)	Handles many clients connecting at once
Advanced	Outgoing connections per second	50 (default 30)	

Choosing the environment and mixed networks

Pick the settings by the **slowest link on the path from the seeder to the switch**. Client speed only affects how fast clients exchange pieces with each other.

Seeder path	Client links	Settings to apply	Note
100 Mbps or slower	100 Mbps or slower	**100 Mbps or slower**	
100 Mbps or slower	1 Gbps or faster	**100 Mbps or slower**	Fast client-to-client exchange greatly shortens total time
1 Gbps or faster	100 Mbps or slower	**1 Gbps or faster**	Lower hardware demand; an HDD can work
1 Gbps or faster	1 Gbps or faster	**1 Gbps or faster**	SSD required

Check both the seeder NIC link speed (`ncpa.cpl` → adapter Status) and the switch uplink spec. The lower of the two decides.

Using the ready-made settings files (qbittorrent.ini)

The `doc\qbittorrent.ini` folder contains qBittorrent settings files with the settings above already applied. Replacing the settings file applies the **Common settings** and the required and recommended items of one environment at once, instead of changing each option by hand.

File	Settings	Notes
`qbittorrent.ini.1G`	Common settings + **1 Gbps or faster**	For a seeder with 8 cores and 16 GB RAM: hashing threads 4, physical memory (RAM) usage limit 4096 MiB; disk queue size left at the default
`qbittorrent.ini.100`	Common settings + **100 Mbps or slower**	Advanced settings at their defaults

Both files are for qBittorrent 5.3.0 (`Meta\MigrationVersion=12` , the settings version of 5.3.0). The lt12 build (libtorrent 1.2) ignores libtorrent 2.x items such as hashing threads, the physical memory limit and the torrent format; with the lt12 build and `qbittorrent.ini.1G` , set **Disk queue size** to 8192 KiB by hand. All other settings, such as window sizes and the language, go back to their defaults.

Location of the settings file

Installation	Folder
Portable version	`<qBittorrent folder>\Profile\qBittorrent\config`
Installed version	`%userprofile%\AppData\Roaming\qBittorrent`

How to apply

1. **Exit qBittorrent completely**, including the tray icon. If the file is replaced while qBittorrent runs, qBittorrent overwrites it with the old settings when it exits.
2. Back up `qBittorrent.ini` in that folder under another name (e.g. `qBittorrent.ini.bak`).
3. Copy the file for your environment into the folder and rename it to `qBittorrent.ini` .
4. Start qBittorrent and **change the items below by hand** to match the seeder PC.

> **Important**
>
> The IP in the files (`203.244.132.147`) is an example. If it is not changed to the seeder's real LAN IP, qBittorrent accepts no connections and the clients cannot reach the tracker.

Menu	Setting	Set to	Why
Tools → Options → Advanced	Network interface	The LAN adapter used for deployment	Adapter IDs differ on every PC, so the file cannot set it

Menu	Setting	Set to	Why
Tools → Options → Advanced	Optional IP address to bind to	The seeder's LAN IP	The file contains an example IP
Tools → Torrent Creator	Tracker URLs	`http://<SEEDER_IP>:9000/announce`	The file contains an example IP

Restart qBittorrent after changing the network interface or the IP. When the execution log (**View → Log**) shows it listening only on the seeder's IP, the binding works. The firewall rules (**Common settings**) and the hardware items of **1 Gbps or faster** (adapter properties, SSD) cannot be set by the file; apply them separately.

Pre-deployment checklist and troubleshooting

Pre-deployment checklist

- [ ] The embedded tracker is on, and `netstat -an | find ":9000"` on the seeder shows LISTENING
- [ ] The firewall has inbound allow rules for the tracker port (9000) and peer port (6881)
- [ ] The torrent was created with Torrent format V1 (lt12 build: Optimize alignment unchecked)
- [ ] The tracker URL IP matches the seeder's current LAN IP
- [ ] The torrent is Seeding, not Queued or Stopped
- [ ] The required items from **100 Mbps or slower** or **1 Gbps or faster** are applied for your environment
- [ ] Sleep, screen saver and automatic update restarts on the seeder will not hit the deployment window

Troubleshooting

Symptom	Cause	Fix
aria2 cannot open the torrent or reports a metadata error	V2 or Hybrid torrent	Recreate it with Torrent format set to V1
Clients cannot reach the tracker	Firewall, wrong IP in the tracker URL, embedded tracker off	Rerun the firewall commands in **Common settings**; check the tracker URL and Enable embedded tracker
A `.pad` folder or empty files appear on clients	lt12 build: Optimize alignment was on	Uncheck it and recreate the torrent
Seeder upload is 0 and no peers connect	Bound NIC or IP is not on the deployment network	Check Network interface and Optional IP address to bind to under Advanced
Stuck at tens of MB/s on 1 Gbps	Upload slot limits, default send buffers, HDD, LAN peer rate limit	Recheck **1 Gbps or faster** values; move the image to an SSD
Every client downloads only from the seeder, never from each other	WinPE firewall blocks inbound client connections	Make sure the client script runs `wpeutil DisableFirewall`

lt12 build (libtorrent 1.2)

The lt12 installer `qbittorrent_5.3.0_lt12_x64_setup.exe` is built with libtorrent 1.2. Everything in this manual applies, except these items:

Page	Setting	Standard build (this manual)	lt12 build
Torrent Creator	Torrent format	Shown; select **V1**	Not shown; torrents are always V1

Page	Setting	Standard build (this manual)	lt12 build
Torrent Creator	Optimize alignment	Not shown	Shown; **uncheck it** (padding files may appear on clients as real files)
Advanced	Hashing threads	Shown	Not available
Advanced	Disk IO type (requires restart)	Shown	Not available
Advanced	Disk queue size (1 Gbps or faster)	Default 102400 KiB; keep it	Default 1024 KiB; set **8192 KiB**
Advanced	Physical memory (RAM) usage limit	Shown	Not available
Advanced	Disk cache / Disk cache expiry interval	Not available	Shown; keep the defaults
Advanced	Coalesce reads & writes	Not available	Shown; keep the default

aria2 cannot download V2 or Hybrid torrents, so with the standard build always check Torrent format; with the lt12 build check Optimize alignment instead.

Choosing the disk to erase

Client local boot

`get_disk.bat` decides which disk is erased, in this order:

1. `FORCE_DISK` in `bats\pe\config.cmd`, if set.
2. The disk whose **disk ID** matches `target_disk.txt` in the image. `make_boot.bat` reads the ID of the disk that holds the running Windows (the GPT disk GUID, or the MBR signature) and writes it into `boot.wim`; WinPE compares it with the `Disk ID` that `diskpart` shows for every disk. Disk numbers can differ between Windows and WinPE, the disk ID does not, and it can be read even when the Windows drive is locked by BitLocker.
3. Only when `AUTO_DISK_FALLBACK=1` (there is no `target_disk.txt`, or no disk has that ID):
    1. the disk of the **only** unformatted (RAW) volume, or
    2. the **only** disk in the PC.

    This fallback is meant for PCs with an **instant recovery program** (a "reboot to restore" program). Such programs can make the Windows partition look RAW in WinPE, and some of them also change what the disk ID looks like from Windows, so the disk ID may not match.

A disk ID on more than one disk (e.g. cloned disks), more than one RAW volume, or more than one disk is treated as an error, so DVT never guesses. Set `AUTO_DISK_FALLBACK=0` to require the disk ID (or `FORCE_DISK`).

> **Note**
>
> The disk ID is compared with the English output of `diskpart`. Keep `PE_LANG=en-us` (the default) in `bats\host\config.cmd`; with another WinPE language the disk ID is not found and only the fallback (step 3) is left.

BitLocker

When BitLocker protects the Windows drive, `make_boot.bat` suspends the protection for **one restart** (`manage-bde -protectors -disable C: -RebootCount 1`). If a check fails in WinPE and the PC returns to Windows, it starts without asking for the recovery key, and Windows turns the protection back on by itself. When the restore runs, the disk is erased anyway. `BITLOCKER_SUSPEND=0` in `bats\host\config.cmd` leaves BitLocker as it is.

The disk ID (step 2 above) can be read on a BitLocker drive, so the disk is found whether the drive is encrypted or not.

USB / CD media

There is no automatic detection. `get_input.bat` shows the disk list and the operator types the disk number (`R` scans the disks again). The disk must exist, and the disk of the boot media cannot be selected. With `MEDIA_CONFIRM=1` (default) the operator confirms with `Y` before the disk is erased.

With `FORCE_DISK` set in `bats\Media\pe\config.cmd` (or in `<media>\config\config.cmd`), that disk is erased and the operator does not choose it: with a static IP only the last octet is entered (e.g. `150`), with DHCP nothing is entered. With `MEDIA_CONFIRM=0` as well, a DHCP boot runs without any input. If the disk does not exist or is the boot media, a warning is shown and the operator selects the disk; answering `N` to the confirmation does the same.

Network

Client local boot

- **DHCP** (default): WinPE waits for an address (`DHCP_WAIT_TRIES` × 2 s) and uses the default route with the lowest metric.
- **Static IP**: if Windows used a static address, `make_boot.bat` stores the address, subnet mask, gateway and the static DNS servers in `ip.txt` inside the image, one value per line:

  ```
  203.244.132.150
  255.255.255.0
  203.244.132.1
  168.126.63.1
  168.126.63.2
  ```

  Lines 4 and 5 (DNS) are written only when Windows has static DNS servers; without them WinPE uses `NET_DNS1/2`. An `ip.txt` with the address only gets `NET_MASK` and the gateway `x.x.x.1`. WinPE tries every connected adapter, keeps the first one that reaches the gateway (and `NET_TEST_HOST`, if set) and returns the others to DHCP.
- **Link speed**: `make_boot.bat` measures it. Links of 1 Gbps or faster get `aria2.conf.1G`; everything else gets `aria2.conf.100`. The console color on the client shows which one was chosen (`COLOR_LINK_*` in `bats\host\config.cmd`).

USB / CD media

The order is set by `DHCP_ENABLE` in `bats\Media\pe\config.cmd`.

DHCP_ENABLE	ip.txt	What happens
`0` or empty (default)	exists	**Static IP first.** The operator enters the last octet and the disk number; the address is applied and the gateway is tested. If the test fails, every adapter is switched to DHCP and the DHCP gateway is tested. When that works, WinPE shows that the static IP failed but DHCP succeeded and continues with DHCP and the disk selected before (`DHCP_FALLBACK_CONFIRM=1` : asks first, `N` stops).
`1`	exists	**DHCP first.** WinPE waits up to `DHCP_FIRST_TRIES` × 2 s (20 s) for a DHCP address and tests its gateway. When both work, only the disk number is asked. Otherwise the static IP is used: the operator enters the last octet and the disk number, and the address is applied and tested.
any	missing	**DHCP only.** Only the disk number is asked.

When no way works, WinPE stops before anything is erased and waits for a key to reboot.

`ip.txt` in `bats\Media\config` has the same format as above, but line 1 is a base address: only its first three octets are used (`ipr`), and the operator types the last one.

```
203.244.132.0
255.255.255.0
203.244.132.1
168.126.63.1
168.126.63.2
```

Line 2 (subnet), line 3 (gateway) and lines 4-5 (DNS) may be omitted (`NET_MASK` , `ipr.1` and `NET_DNS1/2`), so an address-only `ip.txt` also works. Lines starting with `#` are ignored. Save it as ANSI / ASCII, not UTF-8 with BOM.

When the static IP works, a new `ip.txt` with the values actually applied (ip, subnet, gateway, DNS) is written and moved to `T:\temp\ip.txt` for `setpc.bat` . With DHCP (also after the fallback) no `ip.txt` is written.

The link speed is measured in WinPE by `link.exe` , which renames `aria2.conf.1G` or `aria2.conf.100` to `aria2.conf` (`LINK_FALLBACK` when `link.exe` is missing or fails).

Realtek NICs

By default `make_boot.bat` does **not export Realtek network drivers** (`NETDRV_EXCLUDE=Realtek`). Realtek ships **separate drivers for WinPE**, and the regular Windows driver in WinPE can cause errors such as the **connection dropping** during the download. The USB / CD media does not export drivers at all, so it needs the driver in `drivers\` for every NIC that WinPE does not support.

> **Important**
>
> If some PCs have Realtek NICs, put Realtek's **WinPE driver** in the `drivers\` folder of the build PC (e.g. `drivers\realtek\`) and rebuild with `01_make_pe.bat` and `02_make_media.bat` . The drivers in `drivers\` go into every WinPE image.
>
> Emptying `NETDRV_EXCLUDE` to export Realtek's Windows driver is not recommended.

USB / CD media details

config folder

`02_make_media.bat` copies these files from `bats\Media\config` to `<media>\config`, plus `bats\Media\pe\config.cmd`. When WinPE starts, `startnet.cmd` copies `<media>\config` to `X:\restore`, so on a USB drive you can change them without rebuilding. `<media>\config\config.cmd` replaces the `config.cmd` inside `boot.wim`, so its settings (`DHCP_ENABLE`, `MEDIA_CONFIRM`, ...) apply at the next boot. On a CD, change `bats\Media\config` or `bats\Media\pe\config.cmd` and rebuild.

File	Purpose
`DVT_backup.torrent`	required; its `dvt_check.txt` must not be empty (see Preparing the torrent)
`aria2.conf.1G`, `aria2.conf.100`	required; `link.exe` renames the one for the link speed to `aria2.conf`
`ip.txt`	static IP base; tried first or as fallback (`DHCP_ENABLE`)
`config.cmd`	copied from `bats\Media\pe\config.cmd`; WinPE settings

`bats\Media\pe\config.cmd` on the build PC is the original: making or updating a USB drive (`2`, `3`) copies it over a `config.cmd` that was changed on the drive.

USB drive and ISO

- A new USB drive gets one **FAT32** partition (`USB_FS=fat32`), which boots on UEFI and BIOS PCs (BIOS through `bootsect`). The partition size is `USB_PART_MB` in MB; empty (the default) uses the whole disk.
- By default the whole disk is formatted as FAT32 at any size, whatever the Windows build.
- If the format fails on a drive over 32 GB, `02_make_media.bat` shows the diskpart output and a hint. The disk has already been erased at that point. Updating Windows is recommended; otherwise use one of the choices below.
- **If you cannot update Windows, recommended:** make the ISO (`1`) and write it to the USB drive with Rufus. Rufus formats the whole drive as FAT32 at any size, and the drive boots on UEFI and BIOS PCs. In Rufus select the USB drive, select `_output\media\DVT_media.iso`, keep the file system **FAT32**, and write it in ISO image mode. The `config` folder is part of the ISO, so the drive has the same files as one made with `2`; later you can update it with `3`.
- Other choices without a Windows update: `USB_PART_MB=32000` (the rest of the disk stays unused) or `USB_FS=exfat` (whole disk, but many UEFI firmwares do not boot from exFAT).
- `02_make_media.bat` only writes to disks whose type is **USB**, never to the disk that holds `%SystemDrive%`, and asks you to type `YES`.

Behavior in WinPE

- **Reboot loop**: if the PC starts from the USB drive again after the restore, it only stops at the input prompt — nothing is erased without input. The ISO uses the no-prompt UEFI boot sector (`efisys_noprompt.bin`, stored as `efisys.bin`), so a CD left in the drive also boots again without asking — remove the CD while the restore counts down.
- The download drive is chosen automatically (`get_target.bat`): the fixed volume with the most free space, otherwise `T:`. The boot media is never used, even when it is a USB hard disk.

Shared scripts

The WinPE scripts that the client local boot and the media both need are kept once, in `bats\common`. `01_make_pe.bat`, `03_make_test_pe.bat` and `02_make_media.bat` copy `bats\common` into the image after `bats\pe` or `bats\Media\pe`, so a change there reaches every image at the next build. The few differences (the boot media is never a download drive, `pc_ip.txt` is moved instead of copied, the reminder to remove the media) are handled inside the shared scripts: `MEDIA_DRV` is set only by the `startnet.cmd` of the media.

If a script with the same name as one in `bats\common` is still in `bats\pe` or `bats\Media\pe` (left from an older version), the build prints a warning and the `bats\common` copy is used. Delete the old copy.

Configuration

Change values in these files, not in the scripts.

File	Used by
`bats\host\config.cmd`	client PC (`make_boot.bat`) and build PC (`01_make_pe.bat`)
`bats\pe\config.cmd`	WinPE of the client local boot
`bats\Media\config.cmd`	`02_make_media.bat` (loads `bats\host\config.cmd` first)
`bats\Media\pe\config.cmd`	WinPE of the USB / CD media

`bats\host\config.cmd` (client PC and build PC)

Variable	Default	Description
`WORK_DIR`	`C:\temp`	Working folder on the client PC
`BOOT_WIM`	`boot.wim` in the folder of `make_boot.bat` (`C:\temp\boot` or `C:\temp\boot_test`)	WinPE image to customize and boot
`TORRENT`	`%WORK_DIR%\DVT_backup.torrent`	Torrent copied into the image
`NETDRV_EXCLUDE`	`Realtek`	Regex; matching network drivers are not exported. Empty = export all. For Realtek, put the WinPE driver in `drivers\` (see Realtek NICs)
`BOOT_MOUNT_DIR`	`%SystemDrive%\DVT_BootMount`	Local mount folder
`DISK_ID_NAME`	`target_disk.txt`	File in the image that holds the disk ID (must match `DISK_ID_FILE` in `bats\pe\config.cmd`)
`BITLOCKER_SUSPEND`	`1`	`1` = suspend BitLocker on the Windows drive for one restart, `0` = leave it
`BOOT_ENTRY_NAME`	`DVT WinPE`	Name of the boot menu entry
`BCD_CLEANUP_MODE`	`2`	Removal of the boot entry: `1` from WinPE, `2` at the next Windows boot (scheduled task), `0` never
`REBOOT_DELAY_SEC`	`10`	Delay before the restart into WinPE
`ADK_PE_DIR`, `PE_LANG`	ADK amd64, `en-us`	Source of `winpe.wim`

Variable	Default	Description
`BUILD_LOCAL`	`%SystemDrive%\WinPE_LocalBuild`	Local build folder
`PE_SCRATCH_MB`	`512`	WinPE scratch space
`PE_COMMON_DIR`	`bats\common`	Shared WinPE scripts, copied into every image

`bats\pe\config.cmd` (WinPE, client local boot)

Variable	Default	Description
`DL_NAME`	`DVT_backup`	Top folder inside the torrent
`WIM_NAME`	`backup.wim`	Image file inside that folder
`WIM_INDEX`	`1`	Index of the image inside `WIM_NAME` to restore (1 = first image)
`FORCE_DISK`	*(empty)*	Disk number to erase; empty = detect
`DISK_ID_FILE`	`X:\restore\target_disk.txt`	Disk ID written by `make_boot.bat`; checked first
`AUTO_DISK_FALLBACK`	`1`	When the disk ID does not identify the disk, allow the RAW-volume / single-disk fallback (instant recovery programs)
`TDRV`, `SDRV`	`T`, `S`	Letters of the new Windows and system partitions
`EFI_SIZE_MB`, `MSR_SIZE_MB`, `BIOS_SYS_SIZE_MB`	`260`, `16`, `500`	Partition sizes
`NET_MASK`	`255.255.255.0`	Mask for a static IP when `ip.txt` has no subnet line
`NET_DNS1`, `NET_DNS2`	`8.8.8.8`, `8.8.4.4`	DNS for a static IP when `ip.txt` has no DNS lines (4-5)
`NET_TEST_HOST`	*(empty)*	Extra host that must answer (use the tracker or seeder)
`SET_IP_RETRIES`, `SET_IP_RETRY_SEC`	`3`, `10`	Static IP: when no adapter reaches the gateway, all adapters are tried again this many times, after this many seconds
`DHCP_WAIT_TRIES`	`30`	DHCP wait, in 2-second steps
`CHECK_FILE`	`dvt_check.txt`	File used for the seeder check
`SEEDER_TIMEOUT_SEC`	`60`	Seeder check timeout per attempt
`SEEDER_TRIES_1G`, `SEEDER_TRIES_100`	`2`, `5`	Seeder check attempts at 1 Gbps or faster / at 100 Mbps or slower (also when the speed is unknown). `make_boot.bat` writes the link profile to `link_profile.txt` in the image
`SEEDER_EXTRA_OPTS`	*(empty)*	Extra aria2c options for the seeder check
`SPACE_CHECK`	`1`	Disk space check before erasing: `1` = stop when too small, `2` = warn only, `0` = off
`APPLIED_SIZE_MB`	*(empty)*	Size of the applied image in MB. Empty = estimated (`APPLIED_SIZE_PCT`)
`APPLIED_SIZE_PCT`	`200`	Estimated applied size, in percent of the download size

Variable	Default	Description
`SPACE_MARGIN_MB`	`2048`	Space kept free on the Windows partition
`BCD_LOCALE`	`ko-kr`	Locale passed to `bcdboot`
`FAIL_REBOOT_SEC`, `DONE_REBOOT_SEC`	`10`, `5`	Reboot delays

`bats\Media\config.cmd` (02_make_media.bat)

Variable	Default	Description
`MEDIA_LABEL`	`DVT`	Volume label of the ISO and the USB drive
`USB_FS`	`fat32`	File system of a new USB drive: `fat32` or `exfat`
`USB_PART_MB`	*(empty)*	Size of the USB partition in MB; empty = the whole disk
`MEDIA_OUT_DIR`	`_output\media` in the build folder	Result folder (`boot.wim`, `DVT_media.iso`)

`bats\Media\pe\config.cmd` (WinPE, USB / CD media)

It has the same variables as `bats\pe\config.cmd` except `AUTO_DISK_FALLBACK` and `DISK_ID_FILE` (the operator chooses the disk), plus:

Variable	Default	Description
`DHCP_ENABLE`	`0`	`0` / empty = static IP (`ip.txt`) first, `1` = DHCP first
`DHCP_FIRST_TRIES`	`10`	`DHCP_ENABLE=1` with `ip.txt`: 2-second tries for DHCP before the static IP
`DHCP_FALLBACK_CONFIRM`	`0`	Static IP first, static IP failed but DHCP works: `0` = continue with DHCP, `1` = ask Y/N
`LINK_FALLBACK`	`100`	Profile when `link.exe` is missing or fails
`OCTET_MIN`, `OCTET_MAX`	`1`, `254`	Allowed last octet
`FORCE_DISK`	*(empty)*	Disk number to erase; empty = the operator enters it (USB / CD media)
`MEDIA_CONFIRM`	`1`	Ask Y/N before erasing
`DONE_REBOOT_SEC`	`15`	Time to remove the media

Disk space check

`check_space.bat` runs after the seeder check, before anything is erased. It compares the size of the new Windows partition (disk size minus the system partitions and `SPACE_MARGIN_MB`) with what the restore needs:

- the download goes to **another** local drive with enough free space (`get_target.bat`, volumes of the target disk are skipped): the applied image only;
- otherwise the download goes to `T:` itself: the download **and** the applied image.

The applied size is `APPLIED_SIZE_MB` when set. Otherwise it is estimated as `APPLIED_SIZE_PCT` percent of the torrent size (default 200 %, a conservative value for a compressed WIM). For the exact value run `wimlib-imagex info backup.wim <index>` and divide `Total Bytes` by 1048576. If the torrent or disk size cannot be read, the check is skipped with a warning.

aria2 profiles

`bats\pe\aria2.conf.100` and `bats\pe\aria2.conf.1G` (for the media: `bats\Media\config\aria2.conf.*`) are ordinary aria2 configuration files. Both seed for up to 60 minutes after the download, disable DHT/PEX/LPD and run `start_restore.bat` when the download is complete. The 1 Gbps profile allows more peers, removes the upload limit and enables a disk cache.

Version

The DVT version is set in one place, `bats\common\version.cmd` (`DVT_VERSION`). Every `config.cmd` loads it. It is shown in the banner and the window title of the build scripts, `make_boot.bat`, WinPE and the test report.

`restore.bat` writes a record to the registry of the restored Windows:

```
reg query HKLM\SOFTWARE\DVT
```

Value	Example
`Version`	`0.2.0`
`BootMode`	`Local` (client local boot) or `Media`
`RestoredAt`	date and time in WinPE (`%DATE% %TIME%`)
`Torrent`	`DVT_backup`
`ImageFile`, `ImageIndex`, `ImageName`	`backup.wim`, `1`, `Windows 11 Pro`
`Firmware`	`UEFI` or `BIOS`
`IPAddress`	the address used in WinPE
`TargetDisk`	the disk number in WinPE

Script reference

Build PC

Script	Purpose
`01_make_pe.bat`	Builds `_output\boot\boot.wim` from the ADK `winpe.wim` (scripts, tools, drivers, Num Lock on) and copies the host scripts into `_output\boot\`
`03_make_test_pe.bat`	Builds the test image `_output\boot_test\boot.wim` (network, gateway and seeder check, FTP upload of `<ip>.txt`, no disk access)
`02_make_media.bat`	Builds `_output\media\boot.wim` from the ADK `winpe.wim` with `bats\Media\pe`, then writes the ISO, a new USB drive, or updates a USB drive (`iso`, `usb`, `update`)

Client PC (Windows, client local boot)

Script	Purpose
`make_boot.bat`	Main script: prepares the PC and restarts into WinPE (runs from `C:\temp\boot`)
`get_host_ip.bat`	Finds the default-route adapter; writes `ip.txt` (ip, subnet, gateway, static DNS) for a static address only

Script	Purpose
`get_link_speed.bat`	Link speed → aria2 profile and console color
`get_disk_id.bat`	Disk ID (GPT GUID / MBR signature) of the Windows disk
`suspend_bitlocker.bat`	Suspends BitLocker on the Windows drive for one restart
`export_net_drivers.bat`	Exports drivers of connected physical NICs with `pnputil`
`prepare_mount_dir.bat`	Discards a stale mount and checks the mount folder is empty
`set_bcd.bat`	Creates a ramdisk boot entry for `boot.wim`, used for the next boot only
`show_ip.bat`	Shows the last octet of the IP address in large digits

WinPE, shared (`bats\common\`)

Script	Purpose
`version.cmd`	`DVT_VERSION`
`get_firmware.bat`	UEFI or BIOS
`get_ip.bat` / `set_ip.bat`	DHCP / static IP (from `ip.txt`, or from arguments on the media) with DNS
`check_space.bat`	Disk space check before erasing
`check_gateway.bat`	Pings the default gateway
`check_seeder.bat`	Downloads `dvt_check.txt` only and verifies its size (rejects a 0-byte file)
`make_disk.bat`	Cleans the disk and creates GPT or MBR partitions
`get_target.bat`	Picks the download drive (another local volume with enough free space, otherwise `T:`; on the media never the boot media)
`start_restore.bat`	Called by aria2c; starts `restore.bat` once in its own window
`restore.bat`	Applies the image, drivers, boot files, writes `HKLM\SOFTWARE\DVT`, registers `setpc.bat`, cleans up and reboots
`ping_host.bat`, `wait.bat`	Helpers (`timeout.exe` is not available in WinPE)

WinPE, client local boot only (`bats\pe\`)

Script	Purpose
`startnet.cmd`	Runs `start_dvt.bat` when WinPE starts
`start_dvt.bat`	Main flow: checks, network, erase, download
`get_disk.bat`	Disk to erase (disk ID, fallback)

WinPE, USB / CD media only (`bats\Media\pe\`)

Script	Purpose
`startnet.cmd`	Finds the boot media, copies `<media>\config` to `X:\restore`, runs `start_media.bat`
`start_media.bat`	Main flow: network order, input, checks, erase, download
`get_input.bat`	Disk list; last IP octet and disk number (disk from `FORCE_DISK` if set)

Script	Purpose
`getdiskinfo.bat`	Disk list
`link.exe`	Link speed → `aria2.conf`

Troubleshooting

Symptom	What to check
PC reboots into the old Windows after a few seconds	Client local boot: a check failed before erasing. The message on screen names it (disk, network, gateway or seeder).
"The target disk could not be determined"	No disk with the disk ID, and the fallback found no single RAW volume or single disk (or `AUTO_DISK_FALLBACK=0`). Rerun `make_boot.bat`, or set `FORCE_DISK`.
"Disk ID ... was found on more than one disk"	Cloned disks with the same ID. Set `FORCE_DISK`, or remove the extra disk.
"Disk N is too small for this image"	The disk space check failed before erasing. Use a larger disk, put a second disk with free space in the PC, or set `APPLIED_SIZE_MB` (exact size) or `SPACE_CHECK`.
"Image N could not be read from backup.wim"	`WIM_INDEX` does not exist in `backup.wim`. List the images with `wimlib-imagex info backup.wim`.
`make_boot.bat`: "BitLocker could not be suspended"	Suspend it by hand (`manage-bde -protectors -disable C: -RebootCount 1`) or set `BITLOCKER_SUSPEND=0`.
Build: "... is in bats\common and in ..."	A script left from an older version. Delete the copy in `bats\pe` or `bats\Media\pe`.
"No seeder is reachable"	Tracker/seeder running? `dvt_check.txt` in the torrent? Log: `%TEMP%\dvt_seeder.log`.
"The seeder check file in the torrent is empty"	`dvt_check.txt` in the torrent is a 0-byte file. Give it content and create the torrent again.
No network in WinPE	The NIC driver may be missing. Check `NETDRV_EXCLUDE`, or add the driver to `drivers\` and rebuild. Realtek NICs need the WinPE driver.
Connection drops during the download on a Realtek PC	The Windows Realtek driver may be in WinPE. Keep `NETDRV_EXCLUDE=Realtek` and rebuild with Realtek's WinPE driver in `drivers\`.
Media: "The IP settings in ...\config\ip.txt are invalid"	Check the format (ip, subnet, gateway, DNS; one per line) and save `ip.txt` as ANSI / ASCII.
Media: the PC boots the media again after the restore	Remove the USB drive / CD during the countdown. Nothing is erased without input.
`02_make_media.bat` fails to format a large USB drive	FAT32 over 32 GB failed on this build PC. Updating Windows is recommended; otherwise use Rufus, `USB_PART_MB` or `USB_FS=exfat` (see USB drive and ISO).
`make_boot.bat` says the mount folder is not empty	Run `dism /Cleanup-Mountpoints` and empty the folder.
Restore failed after erasing	Read the red message. Fix the cause and run `X:\restore\restore.bat` again from the "DVT Console" window, or press a key to reboot.

A minimized "DVT Console" window is opened in WinPE for troubleshooting. diskpart logs are written to `%TEMP%\dvt_make_disk*.log`.

Limitations

- x64 only (ADK `amd64` WinPE).
- IPv4 only.
- The disk ID is read from the English output of `diskpart`, so WinPE must be English (`PE_LANG=en-us`, the default); otherwise only the fallback is used.
- The disk space check estimates the applied size unless `APPLIED_SIZE_MB` is set.
- The client local boot needs a working Windows on the client PC to run `make_boot.bat`. Use the USB / CD media for PCs without Windows.

Third-party tools

DVT does not include these tools. Download them from their official sites; each is distributed under its own license.

- aria2 — BitTorrent download
- wimlib — applying the WIM image
- NirCmd — stopping aria2c
- Windows ADK and WinPE add-on — Microsoft
- Rufus — optional, writing the ISO to a USB drive
- Veyon — optional, PC management (see above)

License

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The third-party tools above are not part of DVT and keep their own licenses.

Disclaimer

DVT erases disks by design. Use it at your own risk, and test it on hardware you can afford to wipe.